

Amit Shenoy

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LinkedIn: itsamit || *Mission: Solve as many puzzles as possible* || GitHub: xX-its-amit-Xx

SUMMARY

Computational biologist experienced in processing large-scale biological datasets, building reproducible pipelines, and developing statistical models. Skilled in data quality control, troubleshooting workflows, and collaborating with experimental scientists to generate actionable insights.

TECHNICAL SKILLS

Programming: Python, R, SQL (PostgreSQL), Bash

Data Analysis: Statistical modeling, machine learning, Bayesian inference, dimensionality reduction (PCA, UMAP), data visualization, data quality control

Bioinformatics: NGS data processing (FASTQ, BAM), sequencing QC, scRNA-seq analysis, ATAC-seq analysis

Cheminformatics: RDKit, molecular descriptors, ECFP/Morgan fingerprints, QSAR modeling, ADMET prediction

Biophysics: Schrödinger (Maestro, Glide, Prime), docking, MM-GBSA, protein–ligand interaction analysis, VMD, PyMOL

Tools: Git, Docker, Conda, SLURM, Jupyter

EXPERIENCE

UCB Biosciences, Cambridge, MA

2025

Computational Data Science Co-op

- Processed **large multimodal datasets** (chemical, assay, metadata) for drug discovery workflows.
- Developed statistical/ML models to prioritize compounds, reducing screening space to **<1% high-confidence candidates**.
- Identified data artifacts using **UMAP/SHAP**, improving model reliability and interpretability.

COMBINE Lab, Northeastern University

2023–2025

Undergraduate Researcher

- Built **reproducible HPC pipelines (SLURM)** for large-scale molecular modeling and analysis.
- Automated job orchestration and QC workflows, reducing manual intervention by **>80%**.
- Presented findings in **15+ technical talks**; contributed to **2 ongoing manuscripts**.

Arbor Biotechnologies, Cambridge, MA

2024

Gene Therapy Co-op

- Designed **DOE-based experiments** to optimize AAV production, improving system productivity by **>250%**.
- Applied statistical analysis to identify key parameters driving experimental performance.

PROJECTS

OmicsQC & scRNA-Flow (GitHub)

2026

- Built pipelines for sequencing QC and scRNA-seq analysis (filtering, normalization, clustering).
- Implemented metrics to detect low-quality cells, sequencing artifacts, and batch effects.

ATAC-QC

2026

- Developed ATAC-seq QC and peak analysis pipeline with fragment analysis and accessibility profiling.
- Evaluated data quality using TSS enrichment and fragment size distributions.

ARES + BayesBio

2025

- Built ML and Bayesian pipelines for compound prioritization and biological parameter inference.
- Quantified uncertainty and improved decision-making from noisy datasets.

EDUCATION

Northeastern University

Dec 2025

B.S. Bioengineering (Computational, Systems, Synthetic Biology), Minor: Mathematics

GPA: 3.91

Awards: Honors Program; PEAK Research Grants; Dean's List, LEAD360 Emerging Leader

Leadership: STEM tutor; Co-founder, GNU@NU computing community; Co-chair, NU Sustainability; Volunteer.

Interests: Piano, Chess, Tennis, Game Dev, Greek Mythology, Social Deduction Games (BoTC, ONUW), *Death Note*